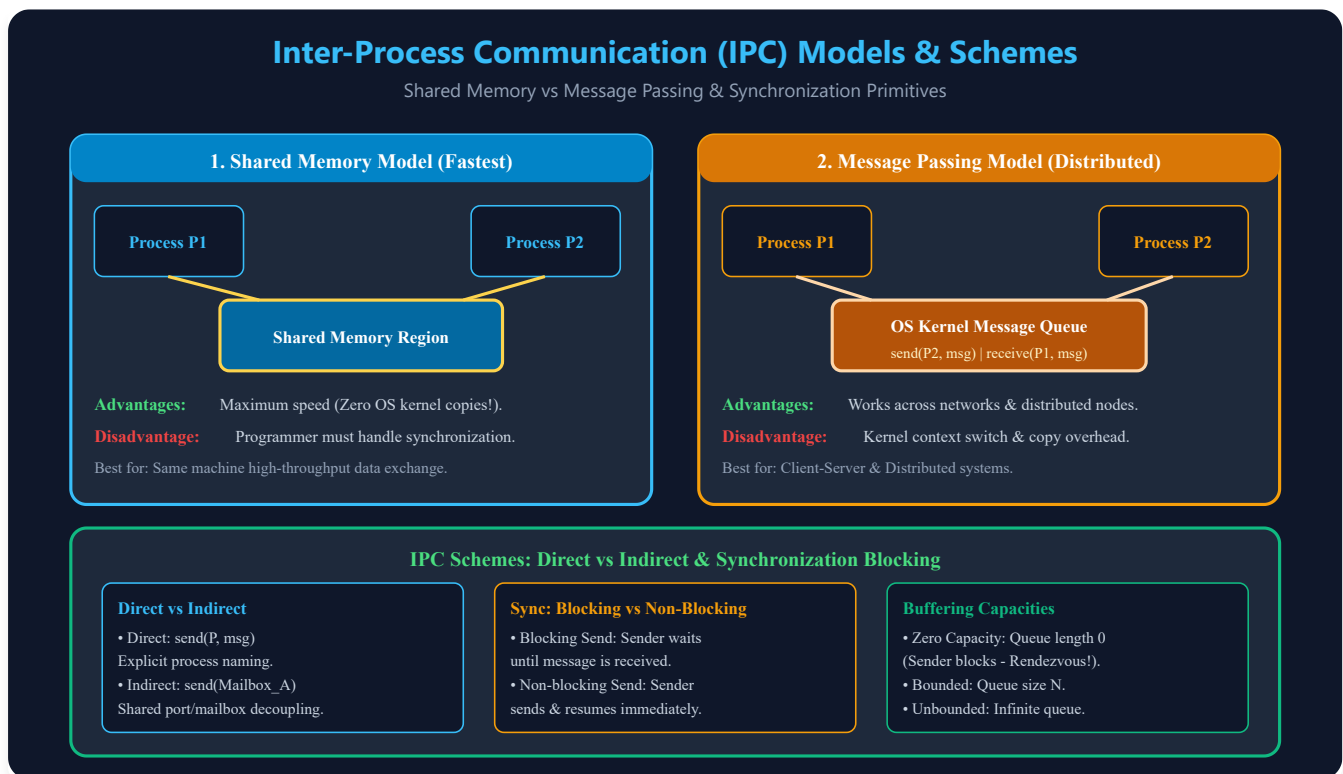


11. Inter-Process Communication (IPC) Models & Schemes

Syllabus Coverage: Gateway Classes Slides 129–137 | Shared Memory vs Message Passing, Direct vs Indirect, Blocking Semantics, Buffering Schemes.

11.1 The Two Fundamental Models

- **Shared Memory:** Fastest mechanism; zero kernel overhead after setup; requires programmer-level semaphore synchronization.
- **Message Passing:** Kernel coordinates message queues; slower due to context switching; ideal for distributed systems.



11.2 Architectural Comparison

Feature	Shared Memory	Message Passing
Communication Path	User memory address space.	Kernel message buffers.
Speed	Extremely Fast (RAM bandwidth).	Slower (Syscalls + data copying).
Synchronization	Manual (Semaphores / Mutexes).	Automatic by kernel.
Network Scalability	Single machine only.	Works across network clusters.